



Submitted Via Email: [REDACTED]
Office of Science and Technology Policy
2415 Eisenhower Ave.
Alexandria, VA 22314

March 15, 2025

Re: Request for Information on the Development of an Artificial Intelligence (AI)
Action Plan, Fed. Reg. No. 2025-02305

To Whom it May Concern:

Luminary Strategies appreciates the opportunity to provide comments on highest priority policy actions that should be in the proposed Artificial Intelligence (AI) Action Plan considered in Federal Register Notice 2025-02305 (Notice). The firm recommends specific work streams in the Action Plan dedicated to identifying and providing national support to several complex and innovative efforts underway in various states to enable power supply development, grid connection, and flexible power consumption technology integration at data center campuses.

The AI revolution requires a modernized, flexible energy procurement and energy usage flexibility framework that fosters competition, encourages private infrastructure investment, and accelerates innovation. Without decisive action, regulatory inertia will continue to hinder the expansion of AI infrastructure, increase costs, and weaken the resilience of the national power grid and the United States' national security imperative for an AI Action Plan.

Statement of Interest

Luminary Strategies represents the interests of:

- AI data center campus developers providing grid-support services and locating near substations with large pockets of transmission congestion,
- Private equity investors in grid infrastructure and power generation assets,
- Battery energy storage developers supporting onsite power, power quality, and grid-to-load stability needs, and
- Software-as-a-service (SaaS) companies building load flexibility tools for AI compute workload management at massive scale to reduce demand from the connected grid while enhancing operational efficiency and resource cost management for the AI campus itself.

These stakeholders are engaged in the development of AI processing campuses and rely on a mix of self-supplied and utility-provided power solutions, regulatory frameworks, and market innovations to achieve efficient, scalable, and sustainable energy management.

Energy for AI – Innovation Discovery Workstream: Identify State-Specific and Utility-Specific Approaches to Inform National Policy

A national strategy for AI energy use must be grounded in real-world solutions emerging from diverse regional, utility, and market-driven initiatives. The federal government's proposed AI Action Plan should include a structured workstream to evaluate and support these developments, focusing on:

1. Hardware and Infrastructure Development

- o Identifying how campus developers and tenants are sourcing and deploying self-supplied equipment versus utility-provided resources.
- o Assessing site-planning strategies that align phased grid connections with multi-stage power demand growth.

2. Software-Driven Load Flexibility

- o Analyzing software innovations that allow data centers to curtail power consumption at peak grid stress periods.
- o Exploring inter-site and inter-campus workload balancing to optimize grid usage and minimize reliance on high-cost peak energy.

3. Regulatory and Market Mechanisms

- o Evaluating utility-driven exceptions, waivers, and incentive programs that accelerate deployment of onsite generation, mobile generation, and wholesale power procurement for large-scale AI processing centers.
- o Examining tariff structures that facilitate cost predictability for data center operators while reducing strain on other energy consumers.

4. Grid Optimization and Infrastructure Investment

- o Investigating best practices for right-sizing grid upgrades, avoiding overbuild, and phasing infrastructure investments to meet AI-related demand.
- o Exploring private-sector-led power delivery models that enhance efficiency and reduce regulatory bottlenecks.

High-Priority Considerations for a National AI Energy Strategy

A comprehensive AI Action Plan must address several critical barriers to ensuring energy resilience and affordability for AI-powered industries.

1. AI Energy Demand Management Solutions are Highly Regional and Fragmented

Innovation in energy supply and demand flexibility is occurring at the state and utility level, often without national coordination. Federal policymakers must identify and standardize best practices to unlock scalable, cost-effective solutions.

2. AI Scalability Hinges on Energy Flexibility

AI infrastructure cannot scale without economies of scale in energy use. Federal government initiatives should promote demand flexibility, workload orchestration, and price-responsive energy management tools to encourage efficient grid utilization.

3. Barriers to Self-Generation and Energy Choice Must Be Addressed

The ability of AI-driven data centers to invest in onsite power generation, storage, and energy efficiency solutions is increasingly constrained by a patchwork of regulatory barriers, restrictive electric codes, permitting hurdles, and opaque utility policies. These limitations not only restrict private infrastructure investment but also create market inefficiencies that drive up costs and reduce resilience.

One particularly problematic constraint is the monopoly control some utilities exert over electricity procurement. In certain regions, local grid providers have fiat authority over commodity power procurement, requiring tenants in a data center campus to purchase power through a full-requirements contract dictated by the utility. This rigid model:

- Prevents cost-effective, competitive procurement of wholesale power, even when cheaper and cleaner alternatives exist.
- Blocks AI campuses from becoming the offtaker on an innovative power purchase agreement (PPA), which could facilitate the financing of new generation capacity to meet their needs.
- Slows deployment of self-generation, battery energy storage, and demand flexibility technologies, despite their proven ability to reduce peak demand stress and improve grid reliability.

Meanwhile, Luminary Strategies' clients across the spectrum of enabling AI dominance through energy procurement and site design are focused on the following efficiencies to grow site viability, attractiveness to AI processing vendors/tenants. The question remains however, how and when these solutions can be a certainty, and not the product of months – or years – of hyperlocal negotiations with incumbent energy commodity and poles-and-wires providers:

- Privately owned substations and distribution infrastructure, ensuring energy delivery reliability while avoiding unnecessary grid overbuild.
- Special Protection Schemes (SPS) and Remedial Action Schemes (RAS) using onsite battery storage to dynamically manage grid constraints, providing fast-track interconnection while utilities implement long-term upgrades.¹
- Onsite renewable energy, battery storage, gas generators, mobile generators, allowing AI campuses to self-supply power to tenants and stagger large interconnection service requests.
- Retail energy structuring, ensuring that self-generation assets and tenants can participate in wholesale energy markets without being locked into restrictive contracts.

A national policy framework should focus on:

- i. Increasing transparency about where these regulatory barriers exist and their impact on AI infrastructure development strategies like those described above.
- ii. Supporting alternative power procurement structures, including private use networks, microgrid campus design, and hybrid retail electricity and power origination models that allow for self-generation, third-party PPAs, and innovative financing mechanisms.
- iii. Encouraging the use of battery energy storage and demand flexibility mechanisms – including software buildout support policies and software testbeds, which can enhance grid stability and reduce the need for costly, long-term transmission investments.
- iv. Creating a standardized process for AI campuses to integrate self-supplied generation and grid interconnection within existing regulatory frameworks, rather than navigating a complex, case-by-case approval process.
- v. Convening on SPS and RAS frameworks as a reliability and speed-to-market solution at

¹ Explanatory Note: Texas ERCOT uses RAS to manage grid constraints by dynamically adjusting power flows during transmission congestion or system disturbances. RAS can curtail loads, adjust generator outputs, or deploy battery energy storage to prevent cascading failures. The Western Electricity Coordinating Council (WECC), covering California, the Pacific Northwest, and the Mountain West, refers to similar measures as Special Protection Systems (SPS). SPS involves automated controls that modify grid operations during contingency events. For example, the Bonneville Power Administration (BPA) has SPS mechanisms that trip generators or curtail load to prevent cascading outages in the Pacific Northwest. Midcontinent ISO (MISO), PJM Interconnection, Southwest Power Pool (SPP), New York ISO (NYISO), and ISO New England (ISO-NE) all implement grid stability programs equivalent to RAS. These regions use a mix of RAS/SPS and curtailment mechanisms, typically enforced by the local Transmission Operators (TOs) rather than a single ISO-wide approach. In PJM and NYISO, large flexible loads (e.g., AI data centers) may negotiate customized demand response agreements or load curtailment plans similar to an ERCOT RAS. □ CAISO does not have an identical RAS but instead utilizes a combination of automated load shedding programs and Flexible Interconnection Transmission Upgrades (FITUs). Large-scale energy users can enter into Curtailment and Load Shedding Agreements that resemble RAS by allowing grid operators to adjust their power use in real time.

federal agencies: FERC and NERC are both overseers on these reliability topics. NERC sets national reliability standards to define how various utility grid operators (system operators) should implement contingency-based corrective actions, and over what timeframe. FERC approves such schemes for the bulk power system for reliability purposes (automatic switching, demand response grid-forming batteries can all be a part of these schemes).

4. A National Dialogue on AI Energy Needs is Essential

U.S. utilities face rate pressures, regulatory scrutiny, and investor concerns over declining returns on investment paired with credit agency downgrades. Utilities as a legacy business model on which most of the U.S. relies to serve 24x7 power needs and support sector-specific innovation, need a national security-focused forum to have open dialogue with large data consumers and tenants to advance national strategy solutions. Without a clear national strategy, energy infrastructure may fail to meet AI's rapidly growing demands. We recommend:

- White House-led summits to foster collaboration among AI firms, energy innovators, and grid operators.
- Where an open forum is not feasible, a closed-door national security-led task force should convene key stakeholders to develop a secure, adaptable energy roadmap for AI infrastructure.

Energy for AI – Key Themes

A robust national policy support agenda will inadvertently lead to consideration of several barriers and themes which have likely been identified in dozens of other comment letters submitted in this proceeding.

1. Energy Consumption and Demand Flexibility Innovation in the United States is Highly Regional, Bespoke, and non-Standardized. National policy can standardize them, if national policymakers know about them.
2. Endless Scale – AI Dominance through Energy Dominance – is Impossible without Economies of Scale, which have to be realized through flexibility plans baked into compute, energy utilization, grid utilization and inter-server orchestration capacities. We need to motivate flexibility – at the interconnection bus, among workloads orchestrating flexible energy consumption through inter-site and inter-campus workload sharing software, and through demand response programs and price signals from utilities and competitive power providers.
3. Americans' right to choose their energy service provider, install their own power generation and storage on their property, install best-in-class equipment to support power reliability, is constrained by a patchwork of regulations, electric code regimes, and bespoke policies that need be unearthed from nontransparent pockets in energy utility companies' customer service programs -this is equally true for data center campuses as it for residential and commercial businesses attempting to installed distributed energy and maximize its onsite utilization and support to the local grid.²

² Comments of the R Street Institute state that "Large energy consumers want the right to choose a competitive energy supplier that has developed faster, lower-cost, and more innovative solutions for datacenters than monopoly utilities. It is important to removing discriminatory regulatory practices, such as restrictions on co-locating power plants and datacenters, while ensuring grid infrastructure costs are generally allocated to those who cause them." Available at: <https://www.rstreet.org/outreach/comments-of-the-r-street-institute-in-request-for-information-on-the-development-of-an-artificial-intelligence-ai-action-plan/>. For at least twenty five years, the same has

Thank you for the opportunity to submit these comments. Please contact the undersigned with any further questions.

Respectfully Submitted,

Arushi Sharma Frank
Principal, Luminary Strategies

already been true for *small* energy consumers: they also want the right to choose, whether through their utility or their competitive energy supplier, faster, lower-cost, more innovative solutions. Even in states where utilities are not in control of power plants and energy choice, control over transmission and distribution lines means that they have fiat authority over who receives power and why, and what you can and cannot install on your personal property.